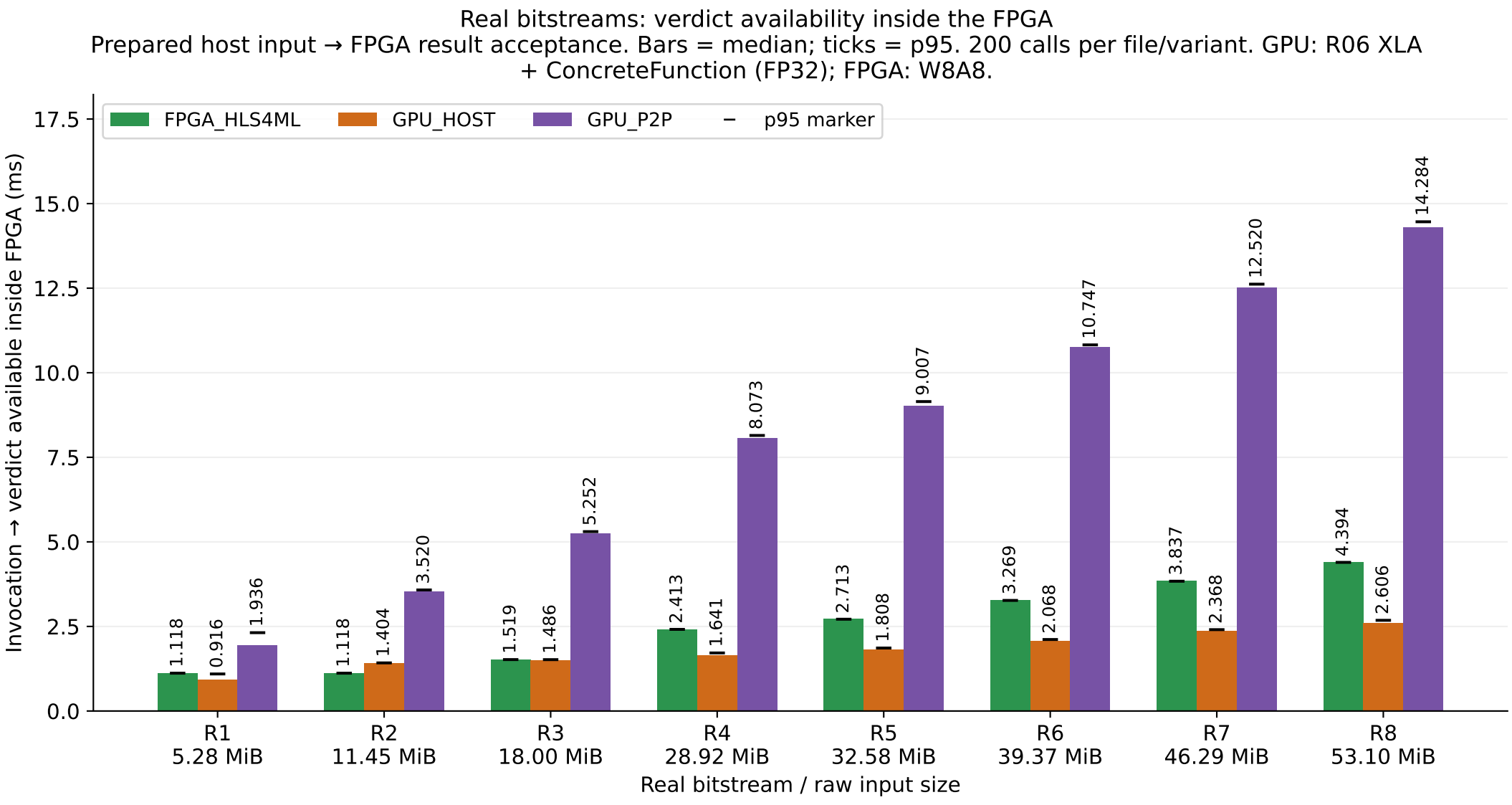
Real bitstreams: verdict availability inside the FPGA
Prepared host input → FPGA result acceptance. Bars = median; ticks = p95. 200 calls per file/variant. GPU: R06 XLA + ConcreteFunction (FP32); FPGA: W8A8.
FPGA_HLS4ML
GPU_HOST
GPU_P2P
p95 marker
Invocation → verdict available inside FPGA (ms)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
1.118
0.916
1.936
1.118
1.404
3.520
1.519
1.486
5.252
2.413
1.641
8.073
2.713
1.808
9.007
3.269
2.068
10.747
3.837
2.368
12.520
4.394
2.606
14.284
R1
5.28 MiB
R2
11.45 MiB
R3
18.00 MiB
R4
28.92 MiB
R5
32.58 MiB
R6
39.37 MiB
R7
46.29 MiB
R8
53.10 MiB
Real bitstream / raw input size